Case Report

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Tapia syndrome after orthognathic surgery: a case report

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Introduction and importance: Tapia syndrome is a cranial neuropathy. It involves the hypoglossal and recurrent laryngeal nerves. The syndrome may follow airway manipulation or prolonged intubation. We report a case of Tapia syndrome, its presentation, management, and outcome.

Presentation: A 45-year-old man with skeletal Class III malocclusion underwent orthognathic surgery. Nasotracheal intubation was difficult and required repeated maneuvers and a second attempt. During extubation, the patient developed laryngospasm and hoarseness, which necessitated re-intubation and 24 hours of mechanical ventilation. After successful re-extubation, he manifested complete tongue immobility and dysarthria. Electromyography performed on postoperative day 10 demonstrated partial right and complete left hypoglossal nerve paralysis. These findings are consistent with Tapia syndrome and correlate with clinical leftward tongue deviation.

Discussion: Management included corticosteroid therapy and initiation of speech and language therapy. We added an adjunctive local low-level laser therapy. Serial clinical photographs and examinations documented steady improvement within 3 months.

Conclusion: Early electrodiagnostic confirmation and multidisciplinary care were associated with near-complete functional recovery by 3 months. The report underscores the importance of careful airway techniques, vigilance for early postoperative bulbar signs, and early rehabilitation to optimize outcomes in iatrogenic cranial nerve injury.

Keywords: case report, corticosteroid therapy, cranial nerve XII palsy, hypoglossal nerve paralysis, hypoglossal nerve paralysis/palsy, laryngospasm, orthognathic surgery, Tapia syndrome

Introduction

Tapia syndrome is an uncommon complication of airway management that usually produces concurrent extracranial injury to the hypoglossal and recurrent laryngeal nerves[1]. The clinical manifestations include tongue weakness, dysarthria, hoarseness, and sometimes dysphagia[1]. Iatrogenic nerve stretch (e.g., due to oropharyngeal packs or malpositioned tubes) or compression during intubation or head positioning is the main mechanism, and the outcomes vary from complete recovery over weeks to persistent deficits in a few cases[2]. The anatomic proximity of the hypoglossal nerve to the lateral tongue base and the recurrent laryngeal/vagal fibers in the hypopharynx explains the vulnerability of these nerves to extrinsic compression, excessive neck extension, or repeated laryngoscopic instrumentation[1]. Moreover, maxillofacial and orthognathic surgeries, with subsequent difficult or prolonged airway manipulation and local bleeding, can contribute to focal neuropraxia[3].

Diagnosis is usually clinical and confirmed or localized with laryngoscopic assessment of vocal-cord function and electromyography when available, while other advanced imaging modalities (e.g., MRI or CT-scan) are usually used to exclude central or structural causes[1]. Management is mostly supportive and multidisciplinary[4]. Accordingly, early speech and swallow rehabilitation is recommended, short empirical corticosteroid courses are commonly used, and several small reports have described adjunctive neuromodulatory approaches such as electroacupuncture or photobiomodulation[5].

HIGHLIGHTS

- This case report documents the Tapia syndrome, manifesting as bilateral hypoglossal nerve palsy with asymmetric electromyography findings, after difficult nasotracheal intubation during orthognathic surgery.
- Multimodal therapy, including corticosteroids, early speech and language therapy, and adjunctive 660 nm low-level laser therapy, facilitated near-complete tongue mobility and speech recovery by postoperative day 90.
- The findings underscore the need to minimize repeated intubation attempts and closely monitor oropharyngeal packing to prevent iatrogenic cranial neuropathies in maxillofacial procedures.

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Prior studies on the Tapia syndrome were mostly focused on using monotherapy and pharmacotherapy, especially corticosteroids. Nevertheless, reports that include multiple approaches for the treatment of the condition, provide clear timelines, and details of airway events are valuable for improving prevention and care. Therefore, this study aims to report on a 45-year-old patient following orthognathic surgery, which can add to the evidence by documenting asymmetric hypoglossal electrophysiology, a predominantly XII-dominant clinical picture without overt vagal palsy, and recovery following bundled rehabilitation and adjunctive interventions. This case report has been prepared in line with the SCARE 2025 checklist[6].

Case presentation

Patient information

A 45-year-old male with a skeletal Class III discrepancy malocclusion, anterior open bite, and maxillary cant was referred to our center. The patient reported no relevant family history of congenital craniofacial anomalies, bleeding disorders, or neurological diseases. He was a non-smoker and did not use alcohol. He had no history of drug allergies or regular medication use. A review of systems was unremarkable except for the chief complaint related to malocclusion and esthetic concerns; specifically, he denied dysphagia, voice changes, neurological symptoms, recent upper respiratory tract infection, or symptoms suggestive of gastroesophageal reflux disease. The treatment plan included maxillary advancement, differential posterior impaction, midline correction with yaw and bodily rotation, bilateral mandibular sagittal split osteotomy, and advancement genioplasty (Fig. 1).

Surgical procedure and postoperative course

He underwent orthognathic surgery in December 2024. Nasotracheal intubation was performed by a board-certified attending anesthesiologist. Maneuvers such as head extension, jaw-thrust, and external laryngeal manipulation were applied to facilitate glottic visualization. Two attempts were made to intubate the patient. The first attempt resulted in massive bleeding due to the presence of large tonsils. Due to difficulty in airway management, a successful intubation was achieved on the second attempt using a Magill forceps and video laryngoscope with a north-facing, 26 cm long, 7.5 mm tube. Intraoperatively, an oropharyngeal pack consisting of a 10 × 10 cm saline-moistened gauze was used. The orthognathic surgery was performed by a consultant oral and maxillofacial surgeon with more than 10 years of experience. The anesthesia

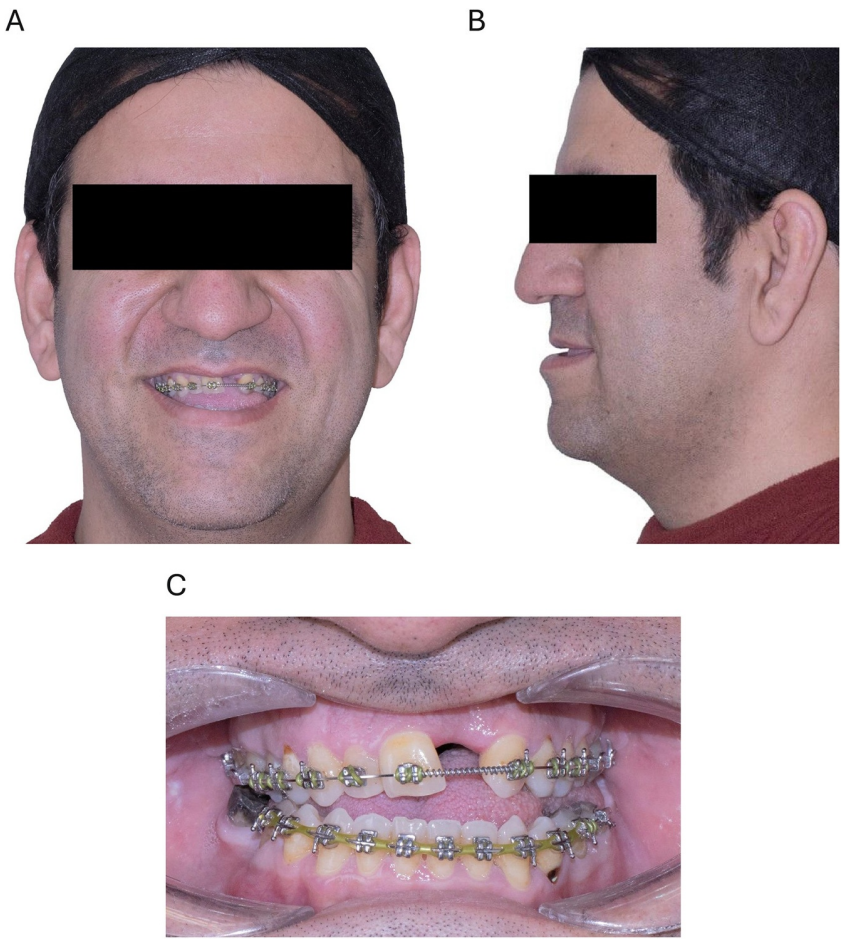


Figure 1. Preoperative records of the 45-year-old patient prior to orthognathic surgery. (A) Frontal facial view; (B) Lateral view; (C) Anterior intraoral view.

protocol included induction with the intravenous hypnotic Propofol (2–2.5 mg/kg) and the opioid Fentanyl (2 mcg/kg), muscle relaxation with the depolarizing agent succinylcholine (1–1.5 mg/kg), and maintenance using the volatile agent Sevoflurane (1.0–2.5% end-tidal concentration) supplemented by an intravenous infusion of the ultra-short-acting opioid Remifentanil (0.1–0.2 mcg/kg/min). The surgical procedure lasted approximately 3.5 hours.

During extubation, the patient developed laryngospasm with hoarseness. Immediate re-intubation was performed by the same attending anesthesiologist, and the patient was transferred to the intensive care unit (ICU). There, he remained mechanically ventilated for 24 hours before successful re-extubation. Subsequent assessment revealed complete tongue immobility and dysarthria. No preoperative or postoperative MRI or CT imaging was performed, as the multidisciplinary team deemed it unnecessary based on the highly characteristic clinical presentation and electromyography findings consistent with neuropraxia; advanced imaging was avoided to minimize radiation exposure, associated costs in a public healthcare setting, and patient burden during the acute recovery phase. Swallowing function remained intact (Fig. 2).

On postoperative day 10, a neurology consultation prompted a cranial nerve evaluation via electromyography, which revealed partial paralysis of the right hypoglossal nerve and complete paralysis of the left, consistent with Tapia syndrome. Clinically, tongue deviation to the left was observed on postoperative day 50 but was absent by postoperative day 90, correlating with the more severe dysfunction of the left hypoglossal nerve (Fig. 3).

Management and outcomes

The patient received three courses of corticosteroid therapy. The first course started immediately after the operation, upon transfer to the ICU. The patient received intravenous dexamethasone on a tapering regimen: 8 mg three times a day on the first day, 8 mg twice a day on the second day, and a single 8 mg dose on the third day, followed by a single stat dose. The second course began on postoperative day 4, after extubation revealed persistent issues. The patient received 8 mg of dexamethasone as a single dose for 3 days. The third course, initiated on postoperative day 14, consisted of oral prednisolone (5 mg) for 1 week.



Figure 2. Initial clinical manifestation of severe lingual paralysis, 10 days post-extubation.

Speech and language therapy began on postoperative day 11. It continued with two sessions per week for 4 weeks. Additionally, adjunctive local low-level laser therapy was administered for 4 weeks with two sessions per week. The laser device used a diode (GaAlAs) laser with a wavelength of 660 nm (red light), an output power of 50 mW, and an energy density (fluence) of 5 J/cm^2 per irradiation point.

Table 1 shows the main clinical events. Serial clinical photographs and examinations from postoperative days 3 to 90 documented progressive recovery (Fig. 3). No complications, such as infection or relapse, occurred, and the patient adhered fully to the treatment plan, with no adverse effects reported.

Discussion

Several procedural and periprocedural features in our patient match commonly reported risk factors. Difficult airway management requiring repeated laryngoscopic maneuvers and a second intubation attempt, intraoperative oropharyngeal packing, bleeding during the first attempt, use of Magill forceps and a north-facing nasotracheal tube, and an episode of laryngospasm with urgent re-intubation are the relevant ones. The nerve injury is, therefore, considered to be the result of a synergistic effect of mechanical trauma during the difficult intubation and prolonged compression from the intraoperative oropharyngeal pack. In this regard, Cariati and colleagues described three iatrogenic Tapia cases after diverse head and neck procedures, in which all patients presented with combined hypoglossal and recurrent laryngeal dysfunction after extubation and achieved near-complete recoveries by approximately three months with conservative management, highlighting the neuropraxic course[7]. Moreover, Izadi *et al* reported Tapia following orthognathic surgery in a younger patient, where prolonged airway manipulation and neck positioning were implicated. This is in accordance with the present report, despite the fact that our patient had bilateral electrophysiologic XII involvement and lacked clinically evident vocal-cord palsy[3]. In another case with nasal surgery, in which oropharyngeal packing and local instrumentation were implicated, the scenario is similar to our intraoperative risk factors (i.e., packing and a transnasal tube orientation), but the mentioned case had more pronounced laryngeal symptoms than the present case, highlighting that local anatomy and the precise site of pressure determine whether X, XII, or both are affected[8].

Although the Tapia syndrome is predominantly attributed to anesthetic factors in orthognathic surgery, a broader multifactorial perspective should consider the procedure's inherent risks, given the anatomical proximity of the surgical field to the hypoglossal nerve pathway. Bilateral sagittal split osteotomy, a common component of mandibular correction, occurs near the lingual aspect of the mandible, where the hypoglossal nerve courses through the sublingual space and could be vulnerable

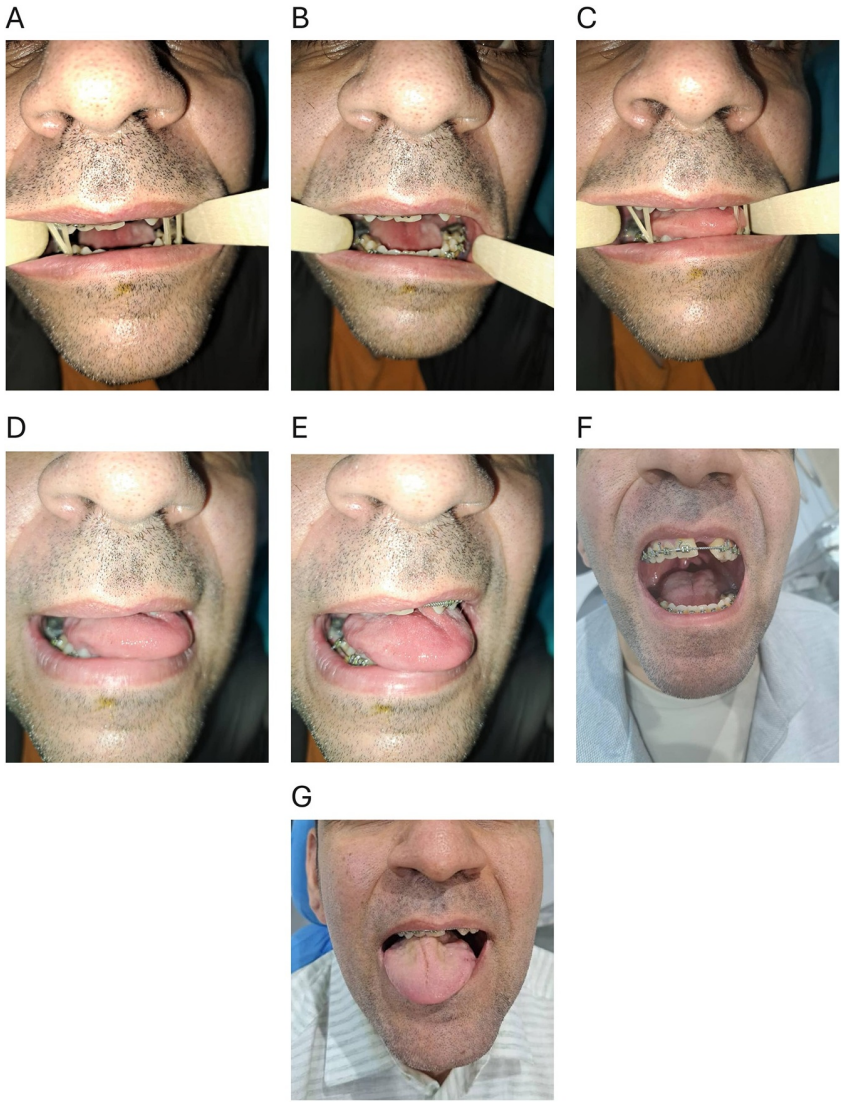


Figure 3. Clinical observations from day 3 to day 90 post-extubation. (A) Day 3; (B) Day 5; (C) Day 10; (D) Day 14; (E) Day 21; (F) Day 50; (G) Day 90.

to indirect compression or stretch during osteotomy, fixation, or soft tissue manipulation[9]. Anatomical variations, such as a longer greater cornu of the hyoid bone, larger hyoid volume, or atypical nerve branching, may exacerbate susceptibility to injury[10]. Additionally, interactions between surgical and anesthetic elements could contribute to neuropraxia in the extralaryngeal space where the hypoglossal and recurrent laryngeal nerves converge[11]. Though the present case lacked imaging evidence of direct surgical trauma, these possibilities underscore the need for intraoperative vigilance, including careful pack placement, optimal tube sizing, and avoidance of excessive neck extension, to reduce iatrogenic cranial neuropathies in orthognathic procedures.

Despite the canonical description of concurrent vagus and hypoglossal involvement, isolated palsies after airway manipulation have been previously reported[12,13]. Hypoglossal injuries can occur as neuropraxic events after transoral or nasotracheal intubation, with clinical manifestations ranging from transient limitation of tongue protrusion to more profound bilateral immobility. The absence of dysphagia or voice disturbance in the present case is therefore somewhat unusual compared to some cases of the Tapia syndrome. It supports the concept that the exact pattern of cranial nerve involvement depends on the precise site, direction, and magnitude of local compression or stretch[12,13].

Regarding the management, empirical corticosteroid treatment is common in prior reports, proposed to reduce local nerve edema and secondary inflammation. In this regard, Cariati's series reported favorable recoveries with conservative care, which is consistent with our three short steroid courses administered early in the postoperative period[7]. In our patient, three steroid courses were administered early in the postoperative period. The clinical timeline (i.e., steady improvement from week 2 onward with near-complete recovery by 3 months) is consistent with other reported neuropraxic recoveries following conservative measures, including steroids[12,13].

Rehabilitation, such as speech and language therapy focusing on tongue strengthening, compensatory articulation strategies,

Table 1

Clinical timeline.

Time point	Event
Preoperative	Referral for skeletal Class III malocclusion; unremarkable family/social history and review of systems.
Day 0 (surgery)	Difficult nasotracheal intubation (two attempts, bleeding); orthognathic procedure completed. Laryngospasm and hoarseness during extubation; immediate re-intubation and ICU transfer with 24-hour ventilation.
Day 1–3	First corticosteroid course (IV dexamethasone tapering regimen).
Day 4	Successful re-extubation; revealed complete tongue immobility and dysarthria. Second corticosteroid course initiated (dexamethasone single dose for 3 days).
Day 10	Neurology consultation; electromyography confirmed bilateral hypoglossal palsy (partial right, complete left).
Day 11	Speech and language therapy initiated (two sessions per week for 4 weeks); adjunctive low-level laser therapy started (two sessions per week for 4 weeks).
Day 14	Third corticosteroid course (oral prednisolone for 1 week). Partial tongue advancement noted.
Day 21	Limited tongue protrusion with leftward deviation; improved speech.
Day 50	Full tongue mobility regained; persistent leftward deviation.
Day 90	Near-complete recovery; no deviation, dysphagia, or voice issues.

and safe swallowing, is another management strategy. Adjunctive modalities, such as electroacupuncture, which was reported in a case study, and photobiomodulation/low-level laser therapy used in peripheral nerve regeneration studies, have growing evidence base for enhancing nerve repair but remain investigational for cranial nerve injuries like Tapia syndrome[1,14]. Accordingly, Xie *et al* reported improvements after electroacupuncture plus swallowing rehabilitation, showing improved deglutition and speech within weeks, which suggests that neuromodulatory or stimulatory therapies may accelerate recovery in selected patients[5]. Moreover, red/near-infrared diode lasers can modulate inflammation and promote axonal regrowth in animal models and some clinical settings, providing a biologic rationale for their use as an adjunct to rehabilitation[1,14]. Mechanisms involve photobiomodulation, where absorbed photons stimulate mitochondrial cytochrome c oxidase, increasing ATP production, reducing reactive oxygen species, and enhancing Schwann cell proliferation and axonal sprouting–effects demonstrated in rodent models of hypoglossal injury[15]. A systematic review of randomized controlled trials has demonstrated the efficacy of LLLT in cranial neuropathies, showing significant improvements in facial disability index scores and paralysis scores when added to standard therapy for facial palsy, with no serious side effects reported[16]. However, limitations include protocol heterogeneity (e.g., variable wavelengths and fluences), absence of sham-controlled trials specific to Tapia syndrome, potential confounding by concurrent therapies, and risks of mild thermal discomfort or inconsistent penetration in deeper parapharyngeal tissues. In our protocol, LLLT was delivered via a GaAlAs diode laser (660 nm, 50 mW) in continuous wave emission mode, targeting five bilateral irradiated points (submandibular triangle, angle of mandible, and upper anterior cervical chain bilaterally, plus one midline suprahyoid point) for a duration of approximately 1.7 minutes (100 s) per point to achieve 5 J/cm^2 fluence, administered non-contact over intact skin.

In terms of prognosis, most previous cases with Tapia syndrome and neuropraxic injuries showed improvement within 8–12 weeks, and many achieved near-complete recovery by three months, while axonotmetic or more extensive injuries portend longer or incomplete recovery[7,13]. The present case showed impairment in the first 2 weeks, followed by gradual improvement with near-normal function by day 90, which is in accordance with prior studies. This supports an overall appropriate prognosis when the lesion is compressive and recognized early.

The strength of the approach in this case lies in the rapid and aggressive initiation of multimodal therapy immediately following the confirmed diagnosis of the Tapia syndrome. The use of prompt high-dose, tapering corticosteroids aimed at reducing perineural inflammation, combined with the early commencement of the speech and language therapy for functional recovery and low-level laser therapy for neuro-biostimulation, provided a comprehensive strategy that likely maximized the potential of the nerve for self-repair. The systematic, serial clinical documentation through follow-up photographs up to 3 months also provided objective evidence of the recovery timeline. However, a limitation might be the absence of pre- or postoperative MRI/CT imaging, which could have provided definitive anatomical confirmation of the lesion in the parapharyngeal space. Furthermore, although a follow-up electromyography at 3–6 months would have provided objective electrophysiological evidence of nerve regeneration to correlate with the excellent clinical outcome, the patient declined repeat testing due to discomfort experienced during the initial study and satisfaction with functional recovery. We recommend serial electrodiagnostic evaluation in future similar cases.

Conclusion

This case report underscores that Tapia syndrome remains an important complication of difficult airway management in orthognathic surgery, particularly where multiple intubation attempts, bleeding, and the subsequent use of mechanical aids and a prolonged nasotracheal tube in a hyper-extended position are involved. The simultaneous onset of hoarseness and lingual paralysis can be the hallmark presentation. Timely, aggressive, and multimodal management, including immediate corticosteroid therapy, speech and language therapy, and low-level laser therapy, facilitates functional recovery. The successful outcome and near-complete resolution of the left-sided tongue paralysis within 90 days, as objectively documented by serial clinical follow-ups, demonstrate the high regenerative capacity of these nerves and highlight the importance of prompt, comprehensive intervention to prevent long-term neurological deficits.

Ethical approval

Ethical approval was not applicable for this case, as the Ethics Committee at Tehran University of Medical Sciences, Tehran, Iran, does not mandate approval for reporting such individual cases.

Consent

Written informed consent was obtained from the patient to publish this report in accordance with the journal's patient consent policy.

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Author contributions

R.M., M.N., and R.S.: conceptualization and study design; R.M., M.N., and R.S: drafting of the manuscript. All authors played a critical role in the revision of the manuscript for important intellectual content. The final manuscript was read and approved by all authors.

Conflicts of interest disclosure

The authors declare that they have no conflicts of interest.

Guarantor

Raheleh Mousavi.

Research registration unique identifying number (UIN)

Not applicable.

Data availability statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

AI-use declaration

We acknowledge the use of ChatGPT-4 for paraphrasing sentences and checking grammar. After using this tool/service, the authors reviewed and edited the content as necessary and take full responsibility for the final content of the publication.

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References

[1] Wei R, De Jesus O. Tapia syndrome. In: StatPearls. Treasure Island (FL): StatPearls Publishing; 2025 January. Accessed 8 June 2024. https://www.ncbi.nlm.nih.gov/books/NBK567747/
[2] Mumtaz S, Henry A, Singh M. Tapia's syndrome. Anesth Prog 2018;65:129–30.
[3] Izadi F, Ahmadi A, Daneshvar A, *et al*. Tapia's syndrome after corrective jaw surgery under general anesthesia: a case report. Iran J Otorhinolaryngol 2017;29:109–11.
[4] Piteira R, Marques D, Carrega F, *et al*. Tapia's syndrome: a comprehensive analysis of a rare intensive care-associated complication. Cureus 2024;16:e53486.
[5] Xie L, Xiong Z, Xiao W, *et al*. Electroacupuncture as an effective therapy for Tapia's syndrome after transoral intubation for general anesthesia: a case report and review of the literature. J Med Case Rep 2022;16:408.
[6] Ahmed K, Ahmed A, Ginimol M, *et al*. Revised Surgical CAse REport (SCARE) Guideline: an update for the age of Artificial Intelligence. Prem J Sci 2025;2. doi:10.70389/PJS.100079
[7] Cariati P, Cabello A, Galvez PP, *et al*. Tapia's syndrome: pathogenetic mechanisms, diagnostic management, and proper treatment: a case series. J Med Case Rep 2016;10:23.
[8] Ma Z, Mao S, Zeng Q, *et al*. Tapia's syndrome following otolaryngologic surgery – a case report. Interdiscip Neurosurg 2020;20:100654.
[9] Sato K, Nasrun NE, Fujita K, *et al*. Unilateral hypoglossal nerve palsy following orthognathic surgery: a case report and literature review. Int J Surg Case Rep 2023;111:108848.
[10] Jawa S, Singh N, Naruka SS. Tapia syndrome following TMJ gap arthroplasty: a case report and review of literature. Natl J Maxillofac Surg 2023;14:511–14.
[11] Fashina AA, Gbotolorun O, Desalu I, *et al*. Tapia's syndrome after surgery for recurrent pleomorphic adenoma of the parotid gland. J Clin Sci 2017;14:148–51.
[12] Leuzinger K, Misra L. Unilateral hypoglossal nerve palsy in a patient with a difficult airway requiring prolonged intubation. Case Rep Anesthesiol 2021;2021:8842503.
[13] Rubio-Nazábal E, Marey-Lopez J, Lopez-Facal S, *et al*. Isolated bilateral paralysis of the hypoglossal nerve after transoral intubation for general anesthesia. Anesthesiology 2002;96:245–47.
[14] Chauhan R, Anushaprasath U, Kaur M, *et al*. Tapia's syndrome: an unanticipated airway complication. Saudi J Anaesth 2022;16:249–50.
[15] Hamblin MR. Mechanisms and mitochondrial redox signaling in photobiomodulation. Photochem Photobiol 2018;94:199–212.
[16] Díaz L, Basualdo J, Chaple-Gil A, *et al*. Effectiveness of low-level laser therapy in patients with maxillofacial neuropathies. A systematic review of randomized controlled trials. Photodiagnosis Photodyn Ther 2025;52:104516.